FORENSEQUENCE : AI-Powered Digital Forensics Investigation System

A PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

Certified that this Project report titled "**FORENSEQUENCE**" is the bonafide work of "**KAVYA SRIRAM (2116241901045), MOHAMED SALMAN FARIS JS (2116241901055)"** who carried out the project under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT

Digital forensic investigations involve analyzing large volumes of evidence from multiple sources, making it challenging to identify suspicious activities efficiently. FORENSEQUENCE is an AI-based digital forensic investigation assistant developed using Python to simplify and organize digital evidence analysis. The system processes log files, files and folders, browser history, and email records to extract relevant information and identify potentially suspicious activities.

It employs a Rule-Based Expert System with Forward Chaining to evaluate extracted facts against predefined IF–THEN rules and generate explainable forensic findings. Weighted risk scoring helps classify suspicious activities, while chronological timeline reconstruction supports the examination of related events. Case management and SHA-256 hashing support evidence organization and integrity verification. SQLite provides local data storage, and automated report generation enables findings to be exported as PDF documents.

The system also incorporates real-time monitoring of selected files and directories to detect relevant changes. By combining evidence analysis, rule-based reasoning, risk assessment, timeline reconstruction, and reporting, FORENSEQUENCE aims to reduce repetitive manual effort and provide a structured approach to digital forensic investigation without relying on machine learning or deep learning algorithms.

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CHAPTER 1
INTRODUCTION

1.1 BACKGROUND

Digital forensics is the process of identifying, collecting, preserving, examining, and analyzing digital evidence to investigate cybercrimes and security incidents. Digital evidence can be obtained from various sources, including system logs, files and folders, browser history, and email records. Analyzing these sources helps investigators understand suspicious activities, reconstruct events, and identify potential security threats.

As the volume and variety of digital evidence increase, examining it manually becomes time-consuming and complex. Evidence from different sources may require different analysis methods, making it difficult to correlate information and identify relevant patterns efficiently. Automated analysis and logical reasoning can help simplify this process by extracting meaningful facts and identifying activities that require further investigation.

FORENSEQUENCE is an AI-based digital forensic investigation assistant developed using Python. It applies a Rule-Based Expert System and Forward Chaining to analyze extracted facts using predefined IF–THEN rules. The system identifies potentially suspicious activities, calculates risk scores, reconstructs chronological event sequences, and generates structured forensic reports. SHA-256 hashing is also used in evidence management to support evidence integrity verification.

1.2 PROBLEM STATEMENT

Digital forensic investigations involve examining large volumes of evidence collected from different sources. Manual examination and separate analysis of evidence formats can increase investigation time and make it difficult to identify suspicious activities efficiently. Furthermore, investigators need a structured way to interpret extracted information, assess potential risks, and understand the sequence of events associated with an incident.

Therefore, a system is needed to integrate evidence analysis, rule-based reasoning, risk assessment, and event timeline reconstruction into a unified workflow. Such a system should

produce understandable findings and organized reports to support digital forensic investigations.

1.3 OBJECTIVES

The main objectives of FORENSEQUENCE are:

- To manage digital forensic cases and evidence files systematically.
- To analyze evidence from logs, files and folders, browser history, and email records.
- To extract relevant forensic facts from different evidence sources.
- To implement a Rule-Based Expert System using predefined IF–THEN rules and Forward Chaining.
- To identify potentially suspicious activities and provide explainable findings.
- To calculate risk scores and reconstruct chronological event timelines.
- To generate structured forensic reports and export investigation results as PDF documents.

1.4 SCOPE

FORENSEQUENCE focuses on assisting digital forensic investigations through automated evidence analysis and rule-based reasoning. The system accepts selected digital evidence sources and processes them to identify relevant facts and potentially suspicious activities. The scope of the project includes:

- Evidence management: Creating investigation cases, managing evidence files, and generating SHA-256 hashes.
- Multi-source analysis: Processing logs, files and folders, browser history, and email data.
- Rule-based reasoning: Applying predefined rules through Forward Chaining to derive findings from extracted facts.
- Risk assessment: Calculating risk scores based on identified activities and their assigned weights.
- Timeline reconstruction: Organizing relevant events in chronological order.
- Report generation: Presenting findings in a structured format and supporting PDF export.

1.5 SECURITY REQUIREMENTS / SECURITY CONSIDERATIONS

Digital forensic applications must handle evidence carefully to preserve its reliability and protect investigation data. FORENSEQUENCE incorporates the following security considerations:

- Evidence integrity: SHA-256 hashing helps verify whether evidence files have changed by comparing their hash values.
- Data confidentiality: Case details and evidence should be accessible only to authorized users.
- Evidence preservation: Original evidence should be preserved, and analysis should be performed without unnecessarily modifying source files.
- Reliable analysis: Evidence parsing and rule execution should produce consistent findings for the same inputs and rule conditions.
- Traceable findings: Rule-based reasoning helps explain why a particular finding was generated.
- Secure data storage: Investigation records and evidence-related information should be stored and handled carefully to reduce unauthorized access or accidental loss.

These considerations support the reliability and responsible handling of digital evidence throughout the investigation process.

1.6 SDG ALIGNMENT OF FORENSEQUENCE

The SDG goals and the corresponding project contribution is shown in the **Table 1.6.**

S. No.	SDG No. & Goal	Relevant SDG Target	Project Contribution / Alignment
1.	SDG 16 – Peace and Justice	Target 16.6 – Effective institutions	Supports transparent digital investigations and reporting.
2.	SDG 9 – Industry and Innovation	Target 9.5 – Technological advancement	Applies automation and rule-based AI to forensic analysis.
3.	SDG 4 – Quality Education	Target 4.4 – Relevant technical and vocational skills	Demonstrates practical digital forensic techniques.

Table 1.6 SDG Goals

CHAPTER 2
LITERATURE REVIEW

2.1 EXISTING SYSTEM WITH SECURITY APPROACHES

Digital forensic investigation involves the examination of digital artifacts to identify activities associated with security incidents. Existing research explores different techniques for evidence analysis, event reconstruction, and logical interpretation. The following studies highlight relevant approaches and challenges in digital forensics.

Hargreaves and Patterson (2012) proposed an Automated Timeline Reconstruction Approach for Digital Forensic Investigations. Their approach analyzes large collections of low-level events, such as file modifications and registry updates, to infer higher-level activities, such as the connection of a USB device. The framework stores extracted events in an SQLite database and identifies patterns to reconstruct meaningful activities. It also preserves the relationship between inferred events and their underlying evidence, helping investigators trace how conclusions were reached. However, reconstructing meaningful events from large volumes of low-level artifacts remains a challenging task.

Bhandari and Jusas (2020) presented an Abstraction-Based Approach for Reconstruction of Timeline in Digital Forensics. Their method organizes forensic events and artifacts into different levels of abstraction, ranging from general activities to specific details. This approach aims to reduce unnecessary information and make timelines easier to understand. The authors evaluated their methodology and reported improvements in timeline reconstruction. However, the study primarily focuses on organizing and interpreting event timelines rather than providing a complete workflow for multi-source forensic analysis and risk assessment.

Breitinger, Studiawan, and Hargreaves (2025) published SoK: Timeline-Based Event Reconstruction for Digital Forensics: Terminology, Methodology, and Current Challenges. Their study reviews existing research on event reconstruction and identifies differences in terminology, methods, and research focus. The authors highlight that previous studies often concentrate on specific aspects, such as timeline generation or tampering detection, resulting in fragmented approaches. They propose a unified framework for understanding timeline-based reconstruction and discuss challenges and future research directions. This work emphasizes the need for more integrated approaches to digital forensic investigation.

Research on expert systems has also explored the use of predefined rules to support digital forensic decisions. A study titled Design of Expert System for Tool Selection in Digital Forensics Investigation describes an IF–THEN rule-based approach for recommending forensic tools based on investigation requirements. The system demonstrates how expert knowledge can be represented as rules to produce structured recommendations. However, its primary purpose is tool selection rather than the direct analysis of evidence and derivation of forensic findings.

Overall, these studies demonstrate the value of automated event reconstruction, structured evidence interpretation, and rule-based decision-making in digital forensics. However, the reviewed approaches focus on different parts of the investigation process, highlighting the importance of integrating evidence analysis, explainable reasoning, risk assessment, and reporting into a consistent workflow.

2.2 RESEARCH GAPS

The literature review identifies several challenges in existing digital forensic approaches. Timeline reconstruction methods help organize digital events, but interpreting evidence from multiple sources remains complex. Rule-based expert systems demonstrate the value of structured decision-making, although their effectiveness depends on the quality and coverage of predefined rules.

The key research gaps identified are:

- Limited integration of evidence from multiple sources into a unified workflow.
- Difficulty interpreting large volumes of forensic artifacts.
- Need for explainable reasoning to connect extracted facts with investigation findings.
- Limited integration of risk scoring with event reconstruction.
- Need for structured reporting that presents findings in an understandable format.

These gaps motivate the development of FORENSEQUENCE, which combines multi-source evidence analysis, rule-based reasoning, risk assessment, timeline reconstruction, and report generation in a single desktop application.

2.3 PROPOSED SYSTEM

FORENSEQUENCE is a Python-based digital forensic investigation assistant designed to support the examination of digital evidence. It combines evidence management, multi-source analysis, rule-based reasoning, risk assessment, timeline reconstruction, and forensic report generation.

The system uses a Rule-Based Expert System with predefined IF–THEN rules. Forward Chaining evaluates extracted facts against these rules to derive findings when their conditions are satisfied. The findings are then used to calculate risk scores and organize relevant events chronologically. The system also supports evidence integrity verification through SHA-256 hashing and presents investigation results in a structured report.

2.3.1 Overview of the Proposed System

The system follows a sequence of evidence collection, preprocessing, fact extraction, rule-based reasoning, risk assessment, timeline reconstruction, and report generation. It processes supported logs, files and folders, browser history, and email records to identify potentially suspicious activities.

Extracted facts are evaluated using predefined rules through Forward Chaining. The resulting findings are used to calculate risk scores and construct chronological timelines. Finally, the system presents the findings and analysis results in a structured forensic report with PDF export support.

2.3.2 Objectives of the Proposed System

The objectives of FORENSEQUENCE are:

- To manage forensic cases and digital evidence systematically.
- To analyze evidence from multiple supported sources.
- To extract relevant forensic facts from digital artifacts.
- To apply IF–THEN rules using Forward Chaining.
- To identify potentially suspicious activities and explain the resulting findings.
- To calculate risk scores and reconstruct event timelines.

- To generate structured forensic reports and PDF documents.

2.3.3 Advantages of the Proposed System

- Unified Analysis: Processes multiple evidence sources through one application.
- Automated Reasoning: Applies predefined rules to derive forensic findings.
- Explainability: Connects findings to the facts and rules that triggered them.
- Risk Assessment: Assigns risk scores to support investigation prioritization.
- Timeline Reconstruction: Organizes events chronologically.
- Evidence Integrity: Uses SHA-256 hashing to support integrity verification.
- Structured Reporting: Presents findings in a readable report with PDF export support.
- Local Processing: Performs rule-based analysis without requiring external AI services.

CHAPTER 3
SYSTEM DESIGN

3.1 HARDWARE AND SOFTWARE REQUIREMENTS

Component	Requirement
Processor	Dual-core processor or higher
RAM	4 GB Minimum
Storage	2 GB free space or more
Operating System	Windows/ Linux
Display	Standard Monitor

Table 3.1.1 Hardware Requirements

The **Table 3.1.1** outlines the minimum hardware specifications required to run the FORENSEQUENCE application

Component	Requirement
python	Core application development
pywebview	Graphical user interface
Pandas	Evidence data processing
SQLite	Local database management
ReportLab	PDF report generation
SHA-256	Evidence integrity verification
Visual Studio Code or another Python IDE	Development environment

Table 3.1.2 Software Requirements

The **Table 3.1.2** lists the software tools and technologies used for application development, evidence processing, data storage, integrity verification, and report generation.

3.2 SYSTEM ARCHITECTURE

FORENSEQUENCE follows a modular architecture designed to process digital evidence and generate meaningful forensic findings. The system is developed using Python and provides a graphical user interface through which investigators can manage cases, upload evidence, analyze suspicious activities, and view investigation results. The architecture consists of an evidence management layer, evidence analysis layer, rule-based reasoning engine, risk and timeline analysis layer, and report generation layer. These components work together to transform raw digital evidence into structured forensic information.

The evidence management component handles case creation, evidence storage, and SHA-256 hash generation. The evidence analysis component processes supported evidence sources, including logs, files and folders, browser history, and email records, to extract relevant facts. The extracted facts are passed to the rule-based reasoning engine, which applies predefined IF–THEN rules using Forward Chaining.

The findings generated by the reasoning engine are used by the risk and timeline analysis component to calculate risk scores and organize events chronologically. Finally, the report generation component presents the investigation results in a structured format and supports PDF export.

System Architecture Flow is visualized in **Fig 3.2**.

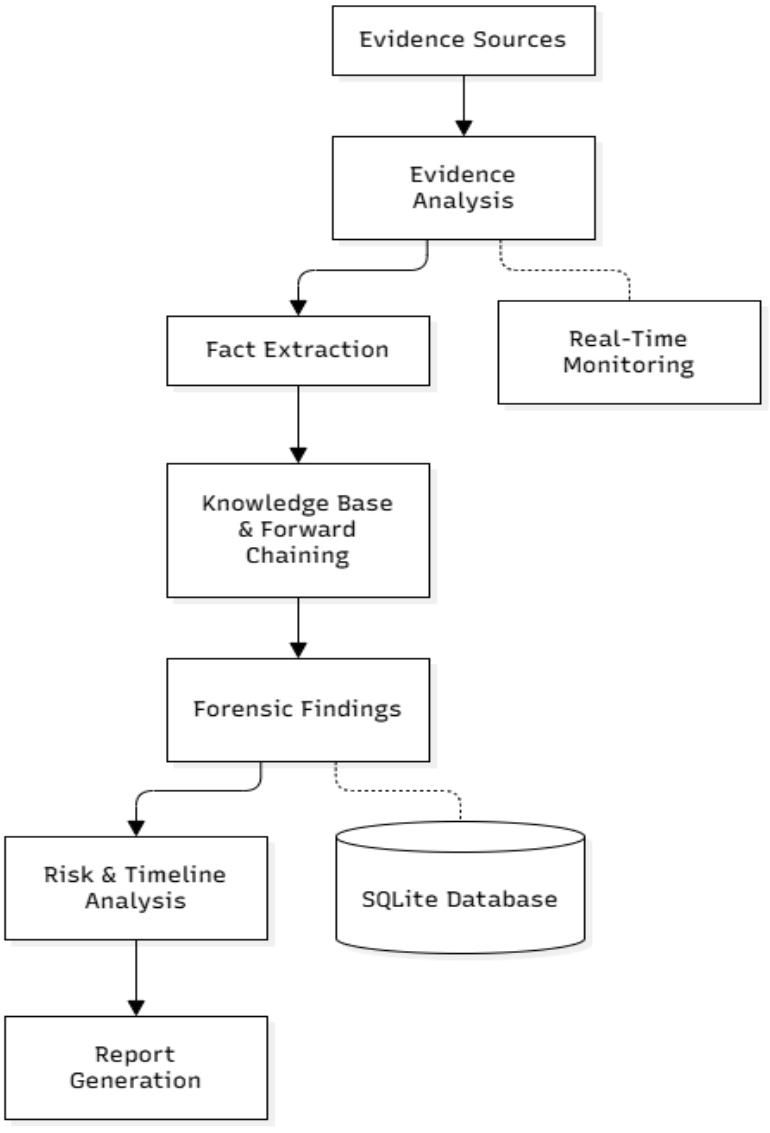


Fig 3.2 System Architecture

3.3 MODULE DESIGN

The system is divided into six functional modules to simplify implementation and organize the investigation workflow.

3.3.1 Case and Evidence Management

This module manages investigation cases and their associated digital evidence. It allows investigators to create cases and upload relevant evidence files for analysis. SHA-256 hashing is used to generate hash values that can help verify evidence integrity. It is shown in the **Fig 3.3.1**.

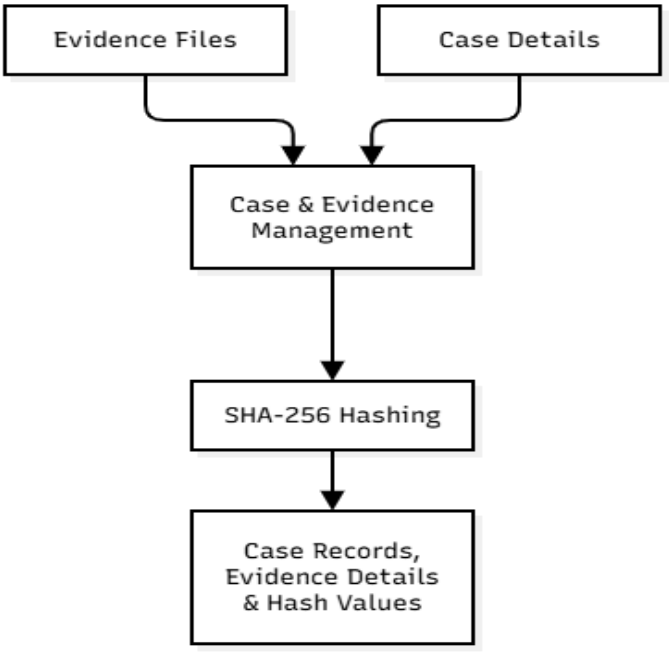


Fig 3.3.1 Case and Evidence Management Flow

3.3.2 Multi-Source Evidence Analysis

This module processes digital evidence obtained from different sources. It supports the examination of log files, files and folders, browser history, and email records. The evidence is parsed to identify relevant information that can be used by the reasoning engine. This is visualized in the **Fig 3.3.2**.

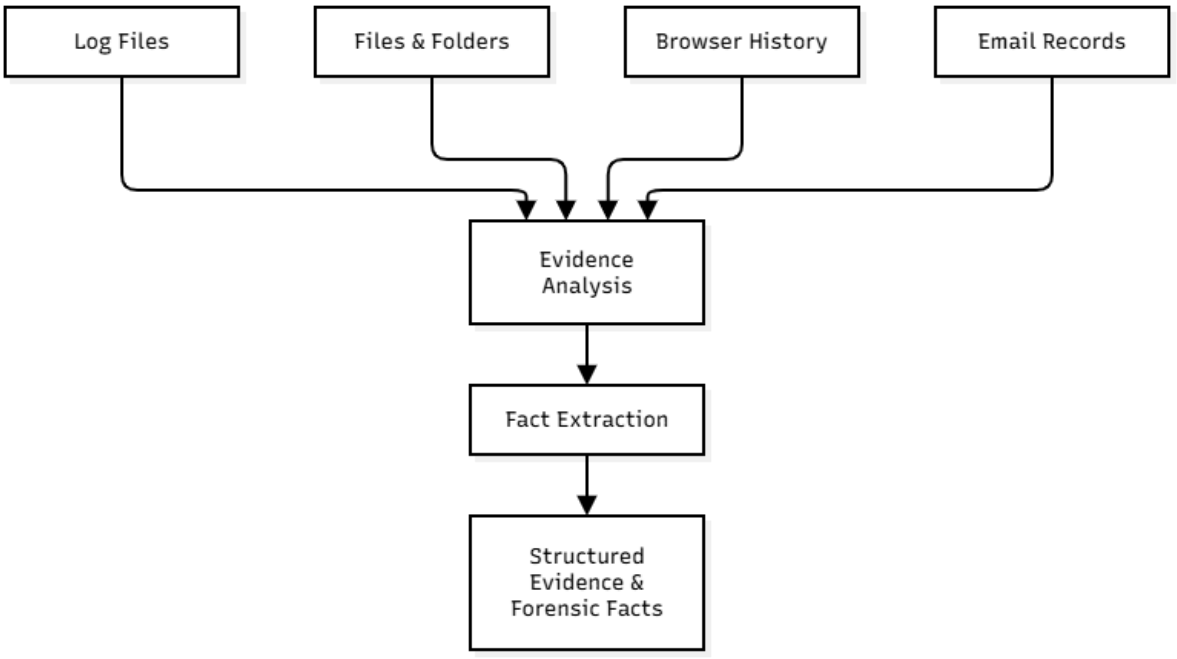


Fig 3.3.2 Multi-Source Evidence Analysis

3.3.3 Rule-Based AI Reasoning

This module identifies potentially suspicious activities using a Rule-Based Expert System. A knowledge base stores predefined IF–THEN rules, while Forward Chaining evaluates extracted facts against the applicable rules and is shown in the **Fig 3.3.3**. When a rule's conditions are satisfied, the system derives the corresponding finding.

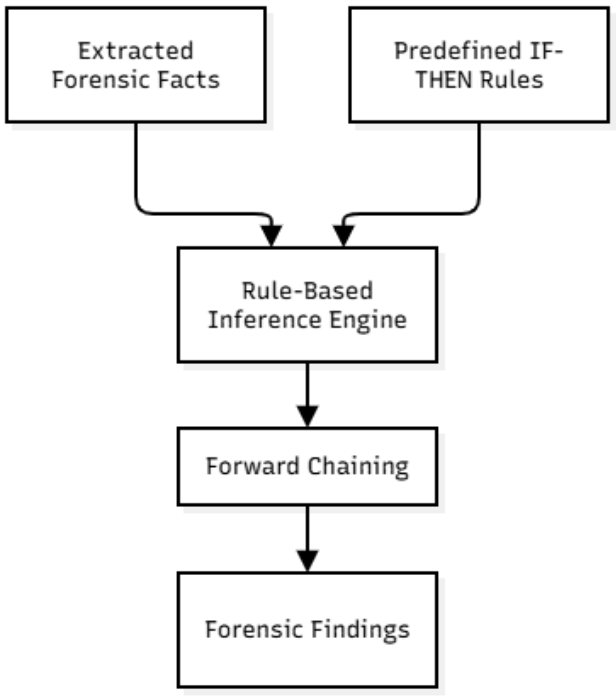


Fig 3.3.3 Rule-Based AI Reasoning

3.3.4 Risk and Event Sequence Analysis

This module evaluates the findings generated by the reasoning engine. Risk weights are assigned to specified findings and combined to calculate a risk score. The score is mapped to a predefined risk level. Relevant events are also arranged chronologically to help investigators understand the sequence of activities. The flow is shown in **Fig 3.3.4**.

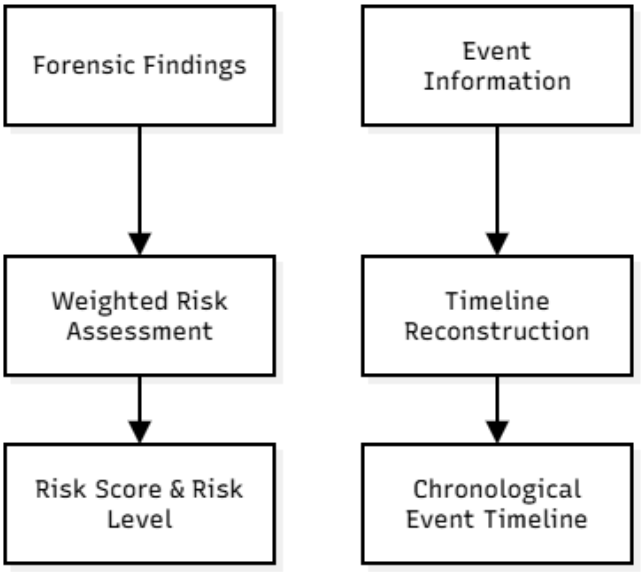


Fig 3.3.4 Risk and Event Sequence Analysis

3.3.5 Forensic Report Generation

This module organizes the investigation results into a readable report. It presents relevant forensic findings, risk information, and event timeline details. The report can also be exported as a PDF document for documentation and review. The flow is shown in the **Fig 3.3.5**.

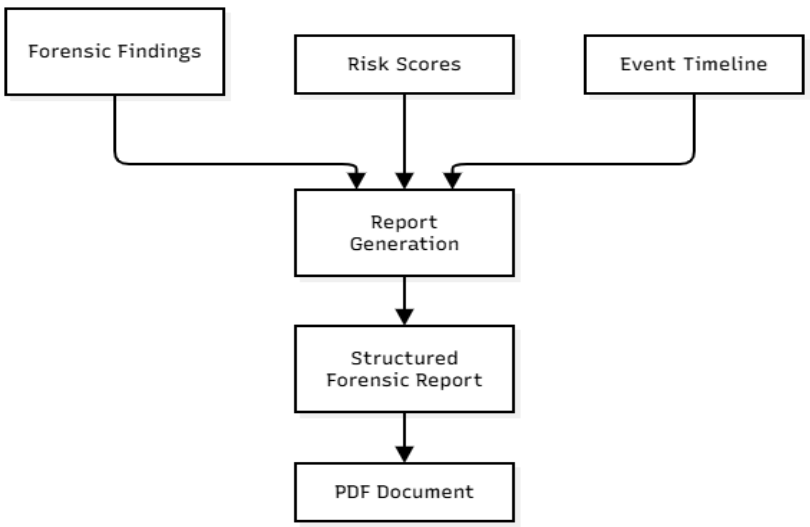


Fig 3.3.5 Forensic Report Generation Flow

3.3.6 Real-Time Evidence Monitoring

This module monitors selected files and directories for changes during an investigation. It detects file-system events such as file creation, modification, and deletion, helping investigators track changes and identify potentially suspicious activities and is visualized in the **Fig 3.3.6**. The detected events can be recorded for further examination.

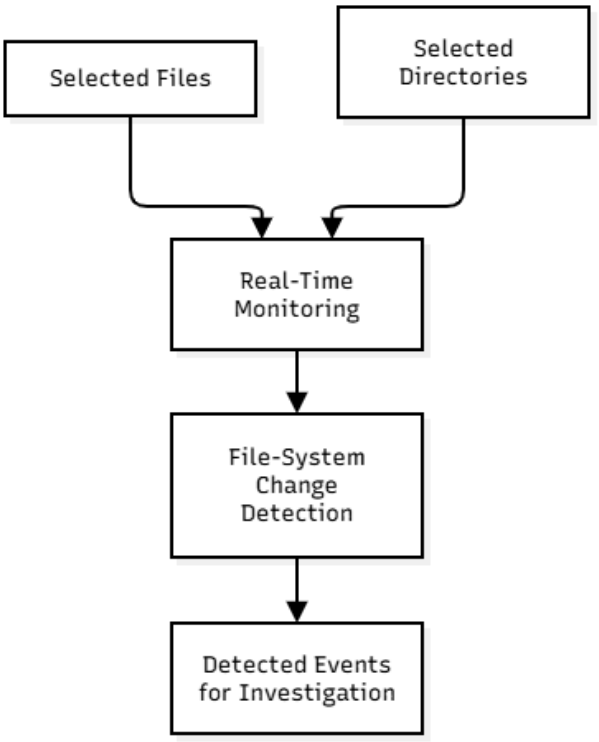


Fig 3.3.6 Real-Time Evidence Monitoring

3.4 UML Diagrams

3.4.1 Use Case Diagram for FORENSEQUENCE

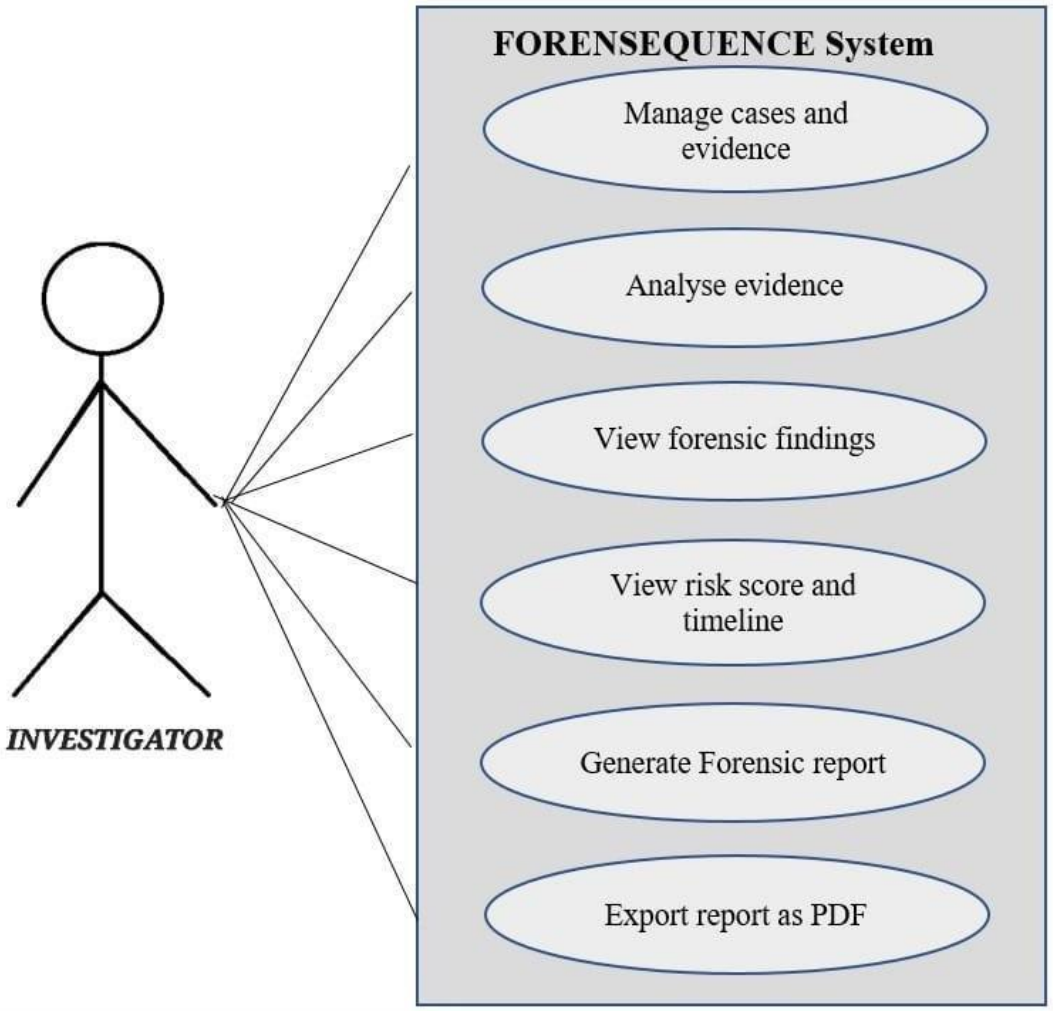
The UML Use Case Diagram in the **Fig 3.4.1** represents the interactions between the investigator and the major functions of FORENSEQUENCE.

The investigator is the primary actor and can perform the following operations:

- Create and manage investigation cases.
- Upload digital evidence.
- Initiate evidence analysis.
- View forensic findings.

- Examine risk scores and event timelines.

- Generate and export forensic reports.



.Fig 3.4.1 Use Case Diagram for FORENSEQUENCE

3.4.2 UML Activity Diagram for FORENSEQUENCE

The Activity Diagram illustrates the sequence of operations performed during a forensic investigation. The workflow begins when the investigator creates a case and uploads evidence. The system processes the evidence and extracts relevant facts. These facts are evaluated using predefined rules, and the resulting findings are passed to the risk assessment and timeline reconstruction components. The system then generates the investigation report and presents the result to the investigator. The diagram is shown in the **Fig 3.4.2**.

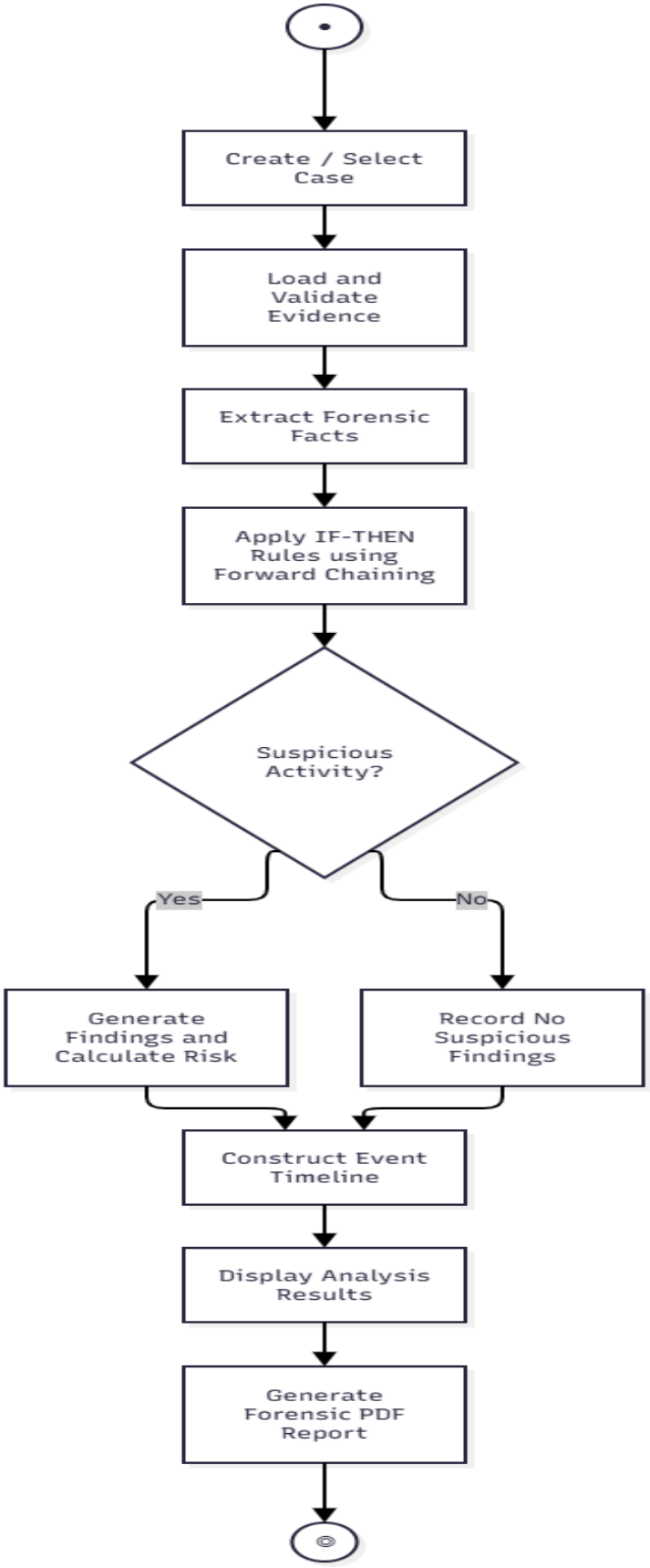


Fig 3.4.2 UML Activity Diagram for FORENSEQUENCE

CHAPTER 4

SYSTEM DESCRIPTION

4.1 DATA COLLECTION

FORENSEQUENCE collects digital evidence from multiple sources for forensic examination. The collected evidence is processed to identify relevant information that may indicate suspicious activities.

The system supports the following evidence sources:

- Log Files: Used to identify failed login attempts and other suspicious system activities.
- Files and Folders: Examined for relevant file information and potentially suspicious artifacts.
- Browser History: Analyzed to identify suspicious domains and browsing-related evidence.
- Email Records: Examined to identify suspicious senders, messages, and executable attachments.

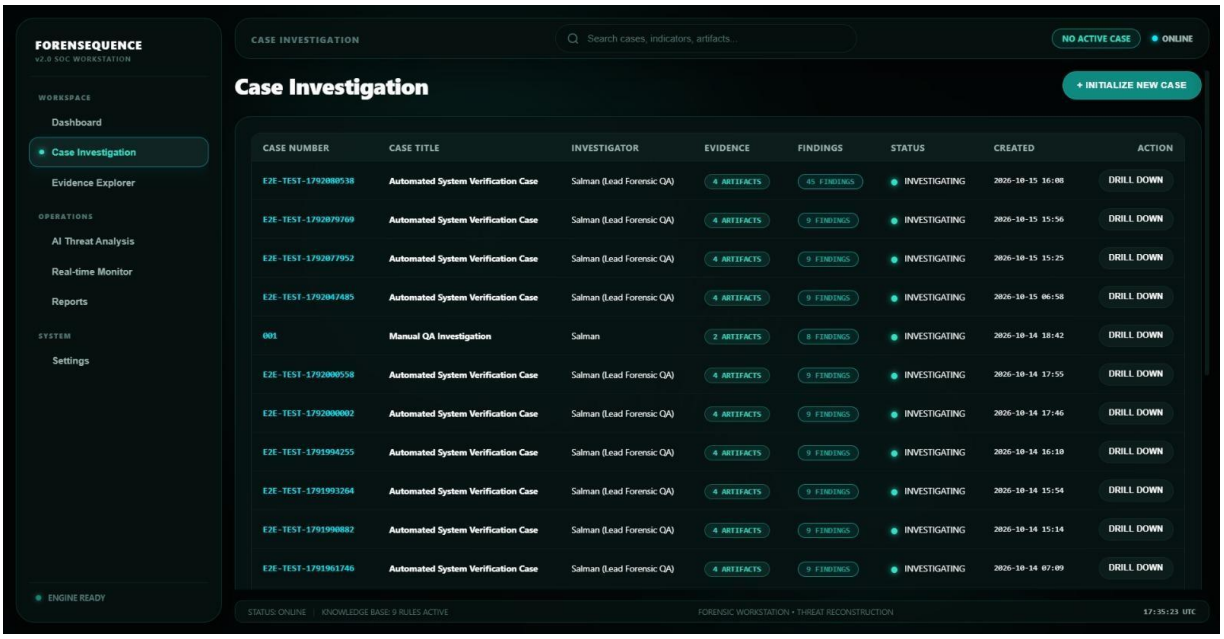


Fig 4.1 Case Investigation and Evidence Management Interface

The Case Investigation and Evidence Management Interface is shown in the **Fig 4.1**. The investigator creates a case and provides the required evidence files through the application. Case details and related information are managed systematically, while SHA-256 hashing supports evidence integrity verification.

4.2 DATA PREPROCESSING

Data preprocessing converts raw evidence into structured information suitable for automated analysis. Since the evidence sources may use different formats, each source is processed according to its supported structure.

The preprocessing steps include:

- Evidence Loading: Read evidence files provided by the investigator.
- Format Handling: Process supported logs, files, browser records, and email data.
- Data Extraction: Identify relevant fields, events, and attributes.
- Fact Generation: Convert extracted information into structured facts for rule evaluation.
- Validation: Handle missing, invalid, or unsupported data during processing.

The extracted facts are passed to the reasoning engine for further analysis.

4.2.1 Algorithms / Methodology

FORENSEQUENCE uses a Rule-Based Expert System with Forward Chaining to identify potentially suspicious activities. The system applies predefined IF–THEN rules to facts extracted during evidence analysis.

4.2.1.1 Rule-Based Expert System

The knowledge base contains rules that represent conditions associated with suspicious activities. For example:

- If failed login attempts exceed a specified threshold, identify suspicious login activity.
- If executable events exceed a specified threshold, flag possible malware-related activity.
- If suspicious domains are detected, flag possible phishing activity.

- If an unknown sender provides an executable attachment, flag a potentially malicious email.

4.2.1.2 Forward Chaining

Forward Chaining is a data-driven reasoning technique that starts with known facts and checks which rules have satisfied conditions. When a rule matches, its conclusion is added to the findings.

The process follows these steps:

- Receive extracted forensic facts.
- Match facts against the rules in the knowledge base.
- Identify rules whose conditions are satisfied.
- Generate the corresponding findings.
- Pass the findings to risk assessment and timeline analysis.

4.2.1.3 Risk Score Calculation

The system assigns predefined weights to selected findings. These weights contribute to a risk score that indicates the potential significance of identified activities. The score is classified into risk levels as shown in the **Table 4.2.1**.

Risk Score	**Risk Level**	**Interpretation**
0-30	Low	Limited suspicious indicators
31-60	Medium	Multiple or significant suspicious indicators
61-100	High	High level of suspicious indicators

Table 4.2.1 Risk Classification Levels

4.2.1.4 Timeline Reconstruction

Relevant event information is arranged chronologically to help investigators understand the sequence of recorded activities and examine their relationship to identified findings.

4.2.2 Results and Discussion

FORENSEQUENCE combines evidence processing, rule-based reasoning, risk assessment, and timeline reconstruction to organize forensic investigation results.

The system extracts relevant facts from supported evidence sources and evaluates them using predefined rules. When a rule's conditions are satisfied, the corresponding forensic finding is generated. Risk scoring helps indicate the relative significance of identified activities, while timeline reconstruction presents relevant events in chronological order.

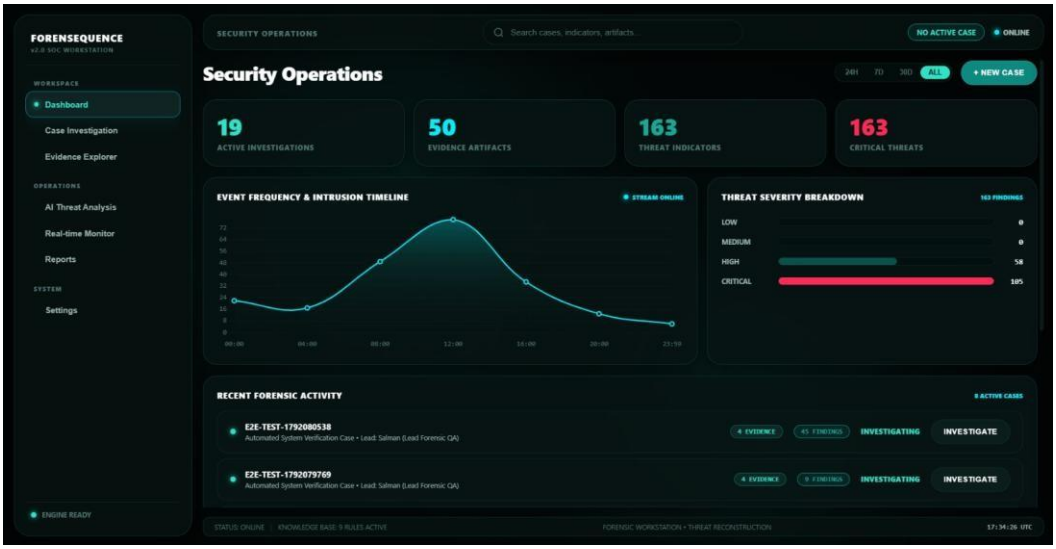


Fig 4.2.2.1 FORENSEQUENCE Dashboard

The **Fig 4.2.2.1** presents the main dashboard, displaying security statistics, threat severity, event trends, and recent forensic activities.

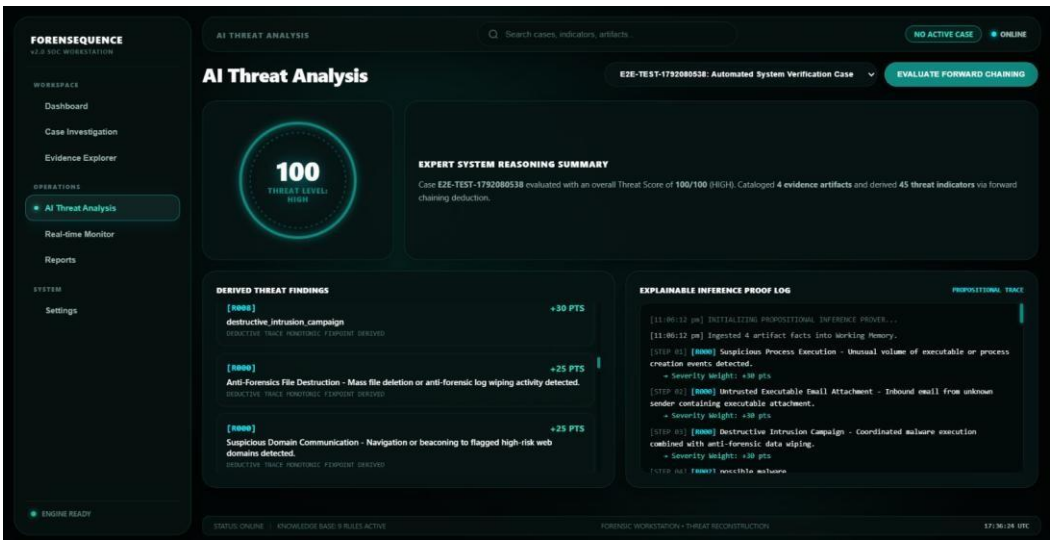


Fig 4.2.2.2 AI-Based Threat Analysis Results

The **Fig 4.2.2.2** illustrates the analysis results, including the risk score, identified threat indicators, and rule-based forensic findings.

The resulting information is organized for presentation to the investigator. The report generation component provides a structured summary of the findings and supports PDF export. The use of predefined rules makes the reasoning process understandable because findings can be associated with the conditions that triggered them. However, the quality of the results depends on the evidence provided, the accuracy of extracted facts, and the coverage of the implemented rules.

4.2.2.1 Functional Testing

Test Case	Expected Output
Evidence file loading	Evidence is loaded for analysis
Rule-based analysis	Applicable forensic findings are generated
Risk assessment	Risk score and level are displayed
Timeline reconstruction	Events are arranged chronologically
PDF report generation	Forensic report is exported as PDF
File-system monitoring	Supported file changes are detected

Table 4.2.2.1 Functional Test Cases

The **Table 4.2.2.1** shows the expected outputs for the corresponding test cases. Overall, FORENSEQUENCE provides a structured approach to digital evidence analysis by combining automated rule evaluation and event timeline reconstruction within a application.

CHAPTER 5

CONCLUSION

FORENSEQUENCE is an AI-based digital forensic investigation assistant developed to simplify the analysis of digital evidence from multiple sources. It integrates evidence management, fact extraction, rule-based reasoning, risk assessment, event timeline reconstruction, and forensic report generation into a single desktop application.

The system uses a Rule-Based Expert System and Forward Chaining to identify potentially suspicious activities through predefined IF–THEN rules. It analyzes log files, files and folders, browser history, and email records to extract relevant information and generate forensic findings. Risk scoring helps classify suspicious activities into different risk levels, while timeline reconstruction organizes events chronologically to support investigation.

SHA-256 hashing supports evidence integrity verification, and SQLite enables local storage of case-related information. The system also supports structured forensic report generation with PDF export. Real-time evidence monitoring extends its capabilities by detecting changes in selected files and directories.

Overall, FORENSEQUENCE demonstrates how classical artificial intelligence techniques can support digital forensic investigations through explainable reasoning and organized evidence analysis. Although its findings depend on the quality of the input evidence and the coverage of predefined rules, the system provides a foundation for further improvements, including expanded rule sets, additional evidence formats, and enhanced investigation capabilities.

CHAPTER 6

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CHAPTER 7

APPENDIX

SYSTEM CONFIGURATION AND TECHNICAL SPECIFICATIONS

7.1 CORE SYSTEM COMPONENTS

Component	Technology	Purpose
Programming Language	Python	Application Logic
User Interface		Desktop interface
Evidence Processing	Pandas and file parsing	Evidence analysis
Rule Engine	Forward Chaining	Derive forensic findings
Integrity Verification	SHA-256	Verify evidence integrity
Database	SQLite	Local data storage
Report Generation	ReportLab	PDF report generation

Table 7.1 System Technologies and Functions

The **Table 7.1** summarizes the core technologies used in FORENSEQUENCE and their respective purposes within the system.

7.2 SYSTEM INTERFACE SPECIFICATIONS

Operation	Input	Output
Case Management	Case details	Case record
Evidence Analysis	Evidence files	Extracted facts and findings
Risk Assessment	Derived findings	Risk score and level
Timeline Reconstruction	Event records	Chronological timeline
Report Generation	Analysis results	Forensic report in PDF format

Table 7.2 Main User Operations

The **Table 7.2** outlines the main system operations, their inputs, and the corresponding outputs generated during forensic investigation.